

## Better management: the unseen opportunity

The IT industry today guzzles power for computers (data centres, desktops, servers, etc) and utilities (air-conditioning, clean water, etc). With the projected increase in IT employment, and the industry in particular, energy consumption will grow exponentially. This makes it imperative that not only green energy but proper energy management be embraced immediately here.

"The unknown opportunity lies in conservation and reducing wastage. One employee switching off the computer monitor after work saves 140W per day; in other words, 40 kW a year. Simple math would show what the impact would be for ten million such employees in India," says Mohan of Distil Information Systems, a company which follows employee-wise carbon footprint calculation and mitigation. That is, every employee's carbon footprint is tracked and double the equivalent amount of credit is ploughed back into the ecosystem. They encourage employees to keep the windows open, reduce unnecessary travel and minimise their carbon footprint, by motivating them through various awards.

Apart from the much-publicised means of energy conservation, there are less obvious means too. "Cloud computing as a concept would help in rationalising server usage across companies, and help in reducing wastage. Better software design by using screen-friendly colours that reduce the electricity consumption, would also help," explains Mohan with examples.

away from the coast. Not only does it become less intrusive, but the effectiveness is also more because the wind speed is usually higher over the ocean than over the land. Denmark, Europe and the UK have been on the forefront in the usage of offshore wind farms. In fact, the UK aims to have enough offshore wind turbines to light up its every home by 2020!

Offshore wind farms are sure to get a fillip from the new deep-water, floating-turbine technologies that are coming up. The first large-capacity floating wind turbine, the Hywind, was constructed and deployed around a year ago. It is a 2.3MW turbine mounted on a 120-metre tall tower in 220-metre deep water in the North Sea off Stavanger, Norway.

Of course, it is not easy to construct sturdy, heavy-duty systems in deep waters, but with robotic submersibles and the precision operation of cranes on the surface, the job becomes doable! If you find this awesome, wait till you hear about the installation of underwater power grids to transmit electricity generated by deep-water wind turbines. An international undersea power grid to connect offshore wind turbines and other ocean energy devices to the land is being proposed in Europe. The of-

ficials believe that having a ready-to-use grid will encourage more people to install such turbines. Right now, Europe has more than 948 offshore wind turbines in 43 fully-operational offshore wind farms, with a total capacity of 2396 MW!

The US had preferred inexpensive on-land wind farms till recently, but realised that most of its population centres are across the coast, and transmission costs and losses would be lower if offshore wind farms were used. Numerous states are now planning offshore wind farms, but there is strong opposition from environmental activists who say it will be harmful for the ecosystem of whales and birds along the coast. So we do not know when it will really happen!

The *Indian Wind Energy Outlook 2009* reports that by 2030, a quarter of India's energy requirements can be met by wind power. Not impossible, considering our 7000km coastline and the currently installed capacity of around 10,000 MW. During a recent visit to India, Steve Sawyer, secretary general, Global Wind Energy Council, commented that India should also look at developing more offshore wind farms. There were some feasibility studies concerning the establishment of an offshore wind farm in Dhanushkoti, near Rameshwaram,

Tamilnadu, but not much was spoken about it thereafter. It might take a few years before India goes for offshore wind farms. Or maybe sooner, considering that China has taken the lead!

## Neighbours, turn your lights on

There are innumerable alternative sources—from solar, wind, hydro, tidal (from waves), biomass, cogeneration (to generate heat and electricity simultaneously), to newer sources like cellulosic ethanol and hot-dry-rock geothermal power to ocean energy. While it is impossible to tap all of these, individuals and governments should take steps to produce energy from as many alternative sources as possible, and manage these better through smart power grids.

"India relies principally on coal for more than half of its total energy generation. However, more than a third of the energy consumed comes from renewable sources, primarily from hydropower. This equation is definitely going to change with the ever dwindling finite source of fossil fuels and the advent of many corporates in the solar segment due to government incentives. These corporates will bring in a scale, which will make the whole segment much more viable," says Hari Kiran, adding that the government of India along with the Ministry of New and Renewable Energy has come up with a roadmap to set up 22,000 MW of solar PV, solar thermal power plants and rooftop solutions by 2020, showing us that the solar industry is going to have an accelerated period of growth in the coming future.

The other major development in this space would be the development of a large number of off-grid power generation units and captive power plants. All this could go a long way in improving the quality of life for every citizen. Let us wait and watch! ●

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